## Recall

### Measures of Central Tendency

It is often convenient to have one number that represent the whole data. Such a number is called a Measures of Central Tendency.

The Measures of Central Tendency usually will be near to the middle value of the data. For a given data there exist several types of measures of central tendencies.

The most common among them are

- Arithmetic Mean
- Median
- Mode

**Thinking Corner**

1. Does the mean, median and mode are same for a given data?
2. What is the difference between the arithmetic mean and average?

**Note**

- Data : The numerical representation of facts is called data.
- Observation : Each entry in the data is called an observation.
- Variable : The quantities which are being considered in a survey are called variables. Variables are generally denoted by $x_i$, $i = 1,2,3,\ldots,n$.
- Frequencies : The number of times, a variable occurs in a given data is called the frequency of that variable. Frequencies are generally denoted as $f_i$, $i = 1,2,3,\ldots,n$.

In this class we have to recall the Arithmetic Mean.

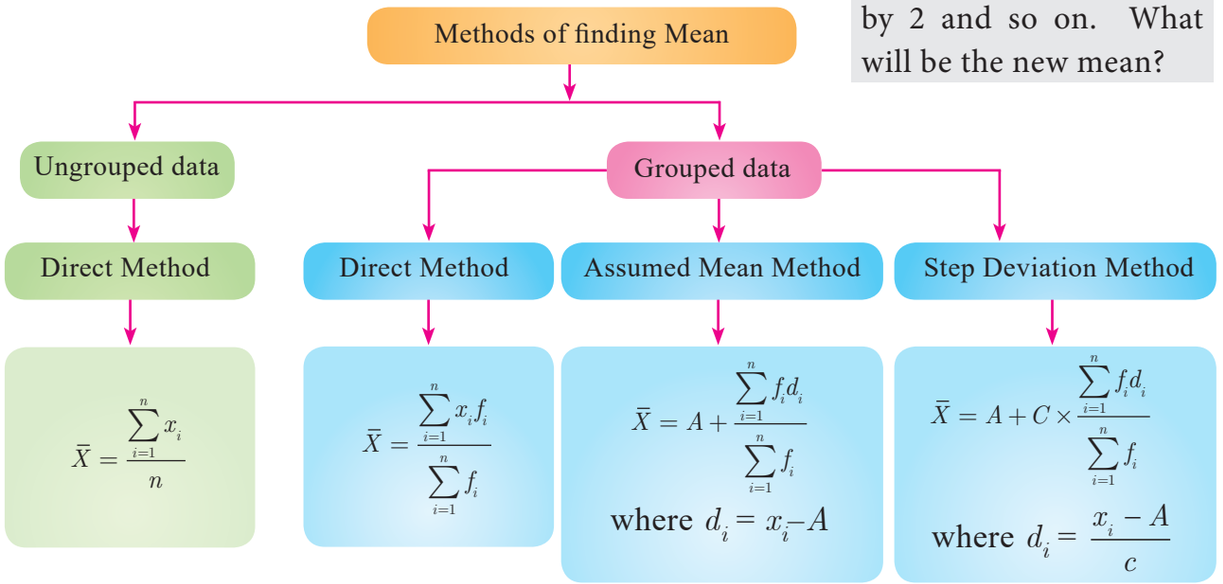
### Arithmetic Mean

The Arithmetic Mean or Mean of the given values is sum of all the observations divided by the total number of observations. It is denoted by $\bar{x}$ (pronounced as $x$ bar)

$$\bar{x} = \frac{\text{Sum of all the observations}}{\text{Number of observations}}$$

**Thinking Corner**

The mean of $n$ observations is $\bar{x}$, if first term is increased by 1 second term is increased by 2 and so on. What will be the new mean?




We apply the respective formulae depending upon the information provided in the problem.



8 Stats and Probability.indd 302 15-12-2020 03:24:04 PM